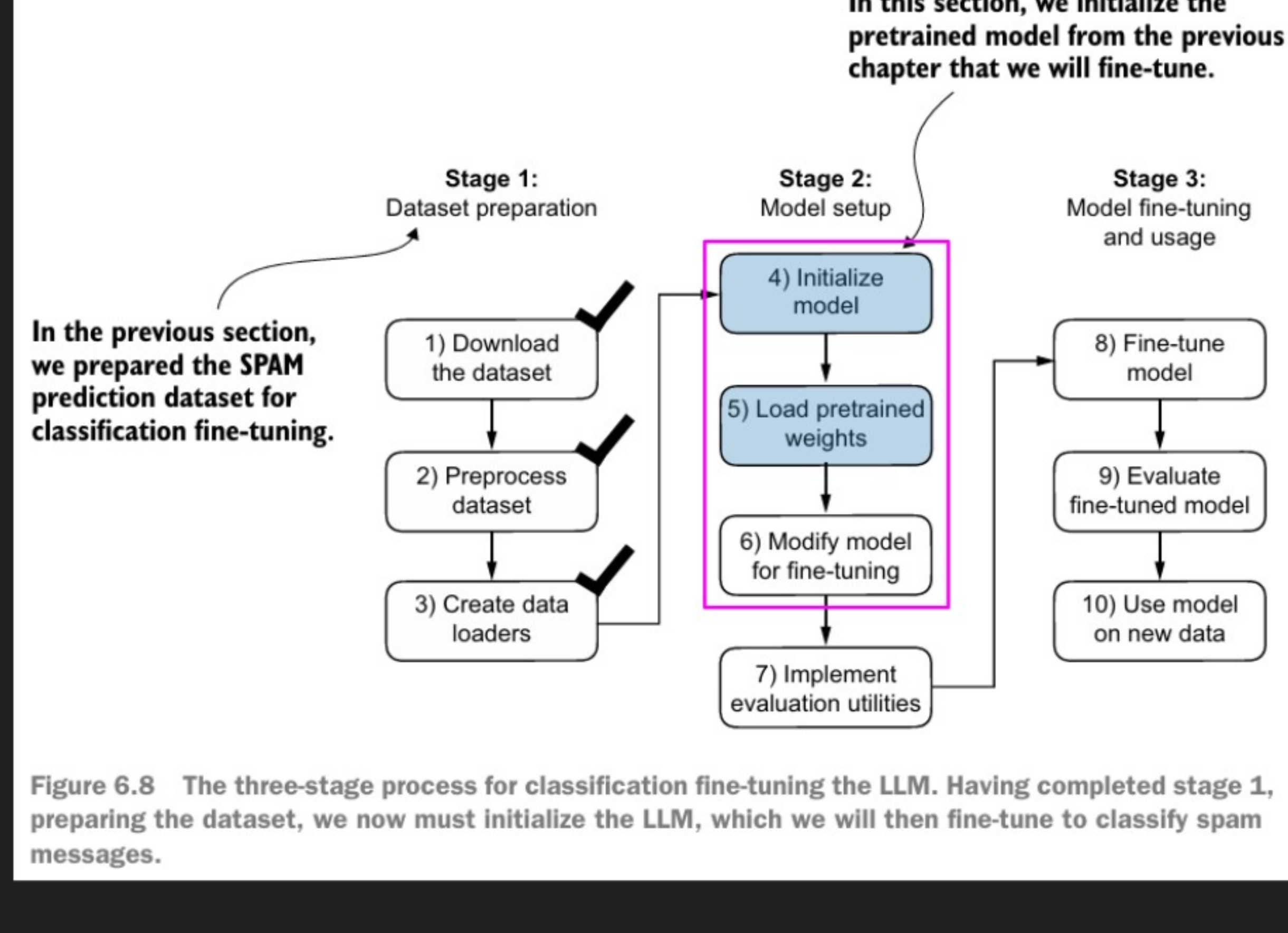
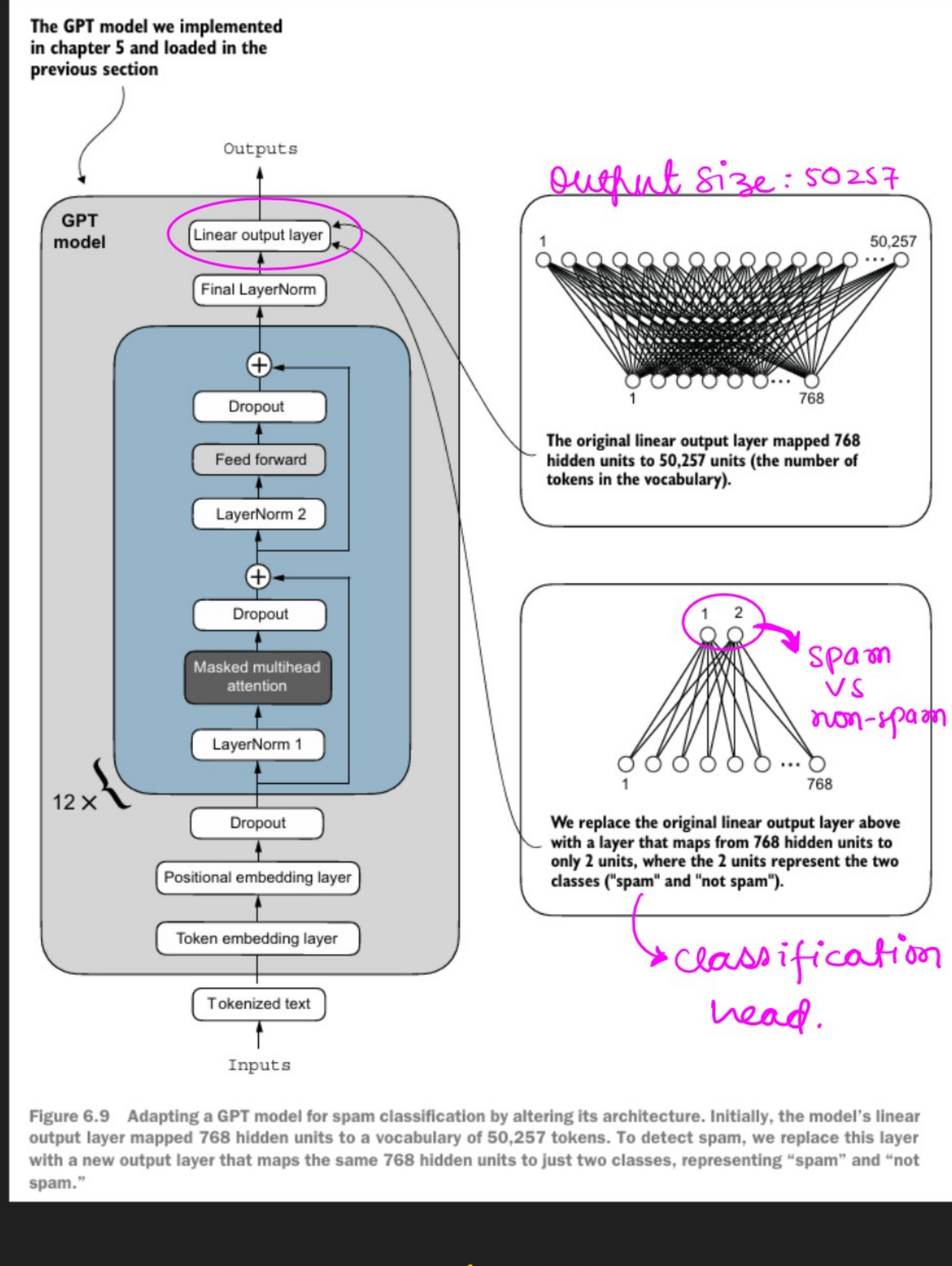


Lecture-34: Model Initialization with pretrained weights

→ In this section, we will prepare the model for classification finetuning.



- Step 1: Load pretrained GPT-2 weight.
- Step 2: Modify the architecture by adding a classification head.



50257 output tensor

every
effort
moves
you

Input Tensor

Here in final row we have to choose token ID with maximum probability.

This is how predict the next token.

- But Now, we don't need to next token prediction. we
- Now we need to just classify the yes or no.

	YES	NO
every	☐	☐
effort	☐	☐
moves	☐	☐
you	☐	☐

Step 3: Select which layers you want to finetune.

Since we start with a pretrained model it is not necessary to finetune all layers.

This is because the lower layers capture basic language structure and semantics, which are applicable across a wide range of tasks and datasets.

Finetuning last layers is enough.

we will finetune:

- Final output head
 - Final transformer block
 - Final layer Norm module
- we will freeze all other parameters.

Step 4: Extract the last output token.

